

RareAgentBench: A Multi-Pillar, Contamination-Aware Benchmark for Rare-Disease Diagnostic Agents

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Abstract

AI agents for rare-disease diagnosis increasingly combine large language models (LLMs) with multi-agent orchestration, retrieval, structured phenotype reasoning, and simulated clinical interaction. Yet published systems are evaluated on incompatible datasets and protocols, making cross-system claims difficult to verify and obscuring whether gains arise from agent scaffolding, model choice, or case selection. We introduce RareAgentBench, an agent-native benchmark that standardizes four capabilities—phenotype extraction, phenotype-only and genotype-aware differential diagnosis, and clinical-communication faithfulness—through a canonical case schema and system-specific adapters for structured HPO and free-text inputs. The benchmark draws on 84,450 cases from five sources, including a time-based PMC holdout, and preregisters 11 hypotheses and 12 ablations. We compare eight diagnostic systems, including classical and offline methods, with three unscaffolded LLM controls. Our evaluation on RareAgentBench yields four key findings: (1) on HPO-input datasets, classical and offline methods outperform all scaffolded LLM agents (best R@1: 0.47 vs. 0.30, a 17-point gap); (2) multi-agent scaffolding does not reliably beat a single-LLM control, changing R@1 by roughly -27 to $+1$ points depending on the scaffold; (3) higher cost does not guarantee higher accuracy—DeepSeek V4-Flash is about $8\times$ cheaper than Gemini Flash but 3–7 points less accurate on HPO inputs, whereas GPT-5 with minimal reasoning costs about $20\times$ more than V4-Flash without a consistent advantage; and (4) structured variant information improves R@1 by about 20 points across systems. We release the benchmark, adapters, execution receipts, and leaderboard to support reproducible and cost-aware evaluation of rare-disease diagnostic agents.

Introduction

Rare-disease diagnosis requires reasoning over heterogeneous evidence, including symptoms, standardized phenotypes, genetic variants, family histories, and free-text clinical narratives. Recent diagnostic systems accordingly extend LLMs with different forms of scaffolding: multi-agent deliberation (MDAgents, MedAgents, and MAI-

DxO), simulated clinical interaction (AgentClinic), retrieval and domain-specific reasoning (DeepRare), structured phenotype-genotype fusion (VC-RDAgent), and phrase mining (RDMA), alongside classical probabilistic approaches such as LIRICAL [1–8]. This rapidly growing design space raises a basic empirical question: which improvements come from the agent architecture, and which come from the underlying model, available evidence, or evaluation set?

Existing results cannot answer this question reliably because the systems are evaluated on different case collections, input representations, and protocols. For example, prior work uses sources ranging from MedQA-Rare and NEJM clinico-pathological cases to proprietary multi-source collections and small HPO subsets. Reported top-1 recall therefore compares neither the same patients nor the same diagnostic task. A recent systematic review of 19 LLM-based rare-disease studies further identified limited preregistration, substantial contamination risk, and a strong association between disease prevalence and reported performance ($R^2 = 0.55$), suggesting that case selection can materially shape apparent gains [9]. Without a shared evaluation infrastructure, claims that one diagnostic agent outperforms another remain difficult to verify.

Benchmarking only the underlying LLM is insufficient. First, the systems differ along non-LLM dimensions—retrieval pipelines, panel orchestration, dialogue policies, and phenotype-genotype fusion—whose effects must be separated from backbone choice. Second, diagnostic accuracy alone omits clinically relevant behavior such as communication faithfulness, run-to-run reliability, and inference cost. Third, classical and offline methods remain strong on structured HPO inputs, but prior agent studies rarely evaluate them alongside LLM-based systems. A useful benchmark must therefore treat the complete diagnostic pipeline, rather than the LLM alone, as the unit of evaluation.

We introduce **RareAgentBench**, an agent-native benchmark designed around this requirement. RareAgentBench evaluates four capabilities in the present study: phenotype extraction, phenotype-only differential diagnosis, genotype-aware differential diagnosis, and faithful clinical communication. It normalizes cases into a shared `Canonical-Case` representation and uses isolated adapters to run heterogeneous systems under a common protocol. Its layered data sources comprise Phenopacket Store (10,051 cases),

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Planned Figure 1: From fragmented evaluation to RareAgentBench

Left: Current practice

Agent A → Dataset A → Metric A
Agent B → Dataset B → Metric B
Agent C → Dataset C → Metric C

Result: reported scores are not directly comparable.

Right: RareAgentBench

Layered sources → CanonicalCase schema
→ system adapters → agents + controls
→ accuracy, faithfulness, reliability, and cost

Result: scaffold, backbone, and input effects are isolated.

Bottom takeaway strip: classical/offline > scaffolded LLMs on HPO inputs | multi-agent gain: none reliable | cost $\nrightarrow$ accuracy

Figure 1: RareAgentBench converts a fragmented evaluation landscape into a controlled comparison. Cases from heterogeneous sources are normalized into a canonical schema and passed through system-specific adapters, allowing diagnostic systems and controls to be evaluated under shared inputs, metrics, and statistical protocols. The figure should emphasize both the source of incomparability in prior work and the benchmark’s main empirical message: scaffold complexity and model cost do not reliably translate into diagnostic accuracy.

RareBench (1,122), RareArena RDS (72,661), a de-leaked MIMIC-IV discharge-note probe (416), and a 200-case time-based PMC Open Access holdout. Together, these layers vary input modality, case prevalence, difficulty, and exposure risk; the PMC layer is drawn after February 2026 to reduce the risk of benchmark contamination. We preregister 11 hypotheses and 12 ablations before holdout unblinding and evaluate accuracy, faithfulness, reliability, and cost rather than collapsing performance into a single score.

We use RareAgentBench to compare eight diagnostic systems, including classical and offline approaches, with three unscaffolded LLM controls. The results challenge several common assumptions. On structured HPO inputs, LIRICAL and offline VC-RD Agent achieve R@1 values of 0.47 and 0.44, respectively, compared with 0.30 for the best scaffolded LLM configuration—a 17-point gap. Multi-agent scaffolding does not reliably improve over single-LLM controls: relative to the no-scaffold control on Phenopacket-Store, the scaffolds range from +0.3 points (MedAgents) down to −27.3 points (MAI-DxO). Backbone trade-offs are similarly non-uniform: DeepSeek V4-Flash is substantially cheaper than Gemini Flash but less accurate, while GPT-5 with minimal reasoning is the most expensive evaluated backbone without a consistent accuracy advantage and degrades sharply in dialogue settings. Finally, adding structured variant information improves R@1 by approximately 20 points across systems, indicating that the gain arises from the information channel rather than a DeepRare-specific mechanism.

Our contributions are fourfold:

- **An agent-native rare-disease benchmark.** RareAgentBench provides layered structured and free-text cases, four evaluated diagnostic capabilities, a canonical case schema, and adapters for eleven methods under a shared protocol.
- **A controlled, preregistered evaluation.** Eleven hypotheses and twelve ablations separate the effects of agent scaffold, LLM backbone, input evidence, and dataset, with bootstrap confidence intervals and multiplicity correction.

- **Cross-system empirical findings.** We show that classical and offline methods can outperform scaffolded LLM agents on structured inputs, that multi-agent gains are modest and dataset dependent, and that model cost does not reliably predict diagnostic accuracy.
- **Reproducible evaluation infrastructure.** We release the benchmark harness, system adapters, 7,581 predictions, per-cell execution and cost receipts, and a static leaderboard to support auditable comparison and future extensions.

Related Work

RareAgentBench lies at the intersection of rare-disease diagnostic AI, LLM-agent evaluation, and agentic systems for medical diagnosis.

Rare-Disease Diagnostic AI and Benchmarks

Rare-disease diagnosis has been studied through both conventional decision-support systems and LLM-oriented benchmarks. LIRICAL ranks phenotype- and genotype-informed hypotheses using interpretable likelihood ratios, while PhenoBrain couples phenotype extraction from electronic health records (EHRs) with disease ranking (Robinson et al. 2020; Mao et al. 2025). RareBench evaluates four rare-disease capabilities of LLMs and investigates knowledge-graph-augmented prompting (Chen et al. 2024); RareArena scales free-text evaluation to screening and confirmation tasks derived from PubMed Central case reports (49,760 screening and 22,901 confirmation cases) (Chen et al. 2026a). Phenopacket Store provides standardized case-level phenotype and genotype records, enabling controlled comparisons with conventional diagnostic software and cross-lingual evaluation (Danis et al. 2025; Reese et al. 2026; Chimirri et al. 2025). On 5,213 phenopacket cases, Exomiser outperformed all seven evaluated LLMs, showing that classical decision support remains an essential baseline rather than a legacy comparator (Reese et al. 2026). More recent resources extend coverage to real EHR notes in MIMIC-RD and to 1,756 multimodal question-answer pairs linked

to 7,958 images in MMRareBench (Eiz AlDin et al. 2026; Ning et al. 2026). Despite broader coverage, these evaluations remain predominantly model-centric: they score static predictions without standardizing agent execution, tool access, interaction, repeated-run reliability, or cost. Nor do they expose an executable interface for applying identical cases and evidence budgets across systems.

Benchmarks for LLM Agents

Agent benchmarks increasingly evaluate interaction trajectories and verifiable outcomes rather than response quality alone. SWE-bench uses tests to verify issue resolution (Jimenez et al. 2024); AgentBoard introduces progress rate for partial credit over multi-step trajectories (Ma et al. 2024); and τ -bench verifies database end states while using pass^k to quantify consistency across repeated trials (Yao et al. 2025). Medical benchmarks adapt these principles to clinical settings. MedAgentBench provides a FHIR-compliant virtual EHR with 300 clinician-authored tasks over 100 patient profiles (Jiang et al. 2025), whereas AgentClinic evaluates tool-using diagnostic interaction with simulated patients across specialties and languages (Schmidgall et al. 2026). MedHELM broadens medical evaluation across task families while considering computational cost (Bedi et al. 2026). These benchmarks motivate trajectory-level, reliability, and efficiency metrics, but none is designed around rare-disease phenotype–genotype reasoning or family-aware diagnosis.

Rare-Disease and Medical Diagnostic Agents

Rare-disease agents employ heterogeneous scaffolds. RareAgents coordinates an LLM-based multidisciplinary team with memory and medical tools (Chen et al. 2026b); DeepRare combines a central host, specialized servers, reflection, and more than 40 tools (Zhao et al. 2026). RDMA focuses on locally mining explicit and implicit rare-disease evidence from EHR notes (Wu, Cross, and Sun 2025), while VC-RDAgent constructs virtual reference cases from structured knowledge bases for offline diagnosis (Liu et al. 2026). Broader medical systems provide complementary designs: MedAgents uses role-based expert discussion (Tang et al. 2024), MDAgents adaptively selects solo or group collaboration (Kim et al. 2024), and MAI-DxO performs sequential, cost-aware diagnosis (Nori et al. 2025). Notably, RDMA targets evidence discovery rather than end-to-end diagnosis, whereas MAI-DxO explicitly models information acquisition and testing cost.

These systems are evaluated on bespoke datasets and protocols that vary in case source, input modality, backbone model, tool access, label normalization, and metrics, precluding controlled comparison. Prior work therefore leaves a three-way gap: rare-disease benchmarks are largely model-centric, agent benchmarks are not rare-disease-specific, and diagnostic agents lack a shared evaluation protocol. RareAgentBench addresses this gap by evaluating complete diagnostic systems through common cases, adapters, and measures of accuracy, faithfulness, reliability, and cost.

Benchmark Design

RareAgentBench treats the complete diagnostic system as the unit of evaluation. Its design combines a clinically motivated capability model, a layered data stack, a canonical case representation, and a controlled protocol that separates downstream reasoning from end-to-end performance.

Evaluation Capabilities

We define five distinct but complementary capability pillars following the diagnostic workflow from phenotype recognition to evidence-grounded communication. The present study evaluates four pillars; family-aware diagnosis is retained in the schema and design matrix but deferred until sufficiently large pedigree-bearing datasets are available.

P1: Phenotype extraction maps free-text clinical narratives to ranked Human Phenotype Ontology (HPO) terms and is evaluated using precision, recall, and F1. **P2: Phenotype-only differential diagnosis** maps curated HPO terms to a ranked disease list and is evaluated using $\text{Recall}@1/3/5/10$, mean reciprocal rank (MRR), and median rank. **P3: Genotype-aware differential diagnosis** adds structured variants and requires both a disease prediction and a causative-gene hypothesis; metrics include disease $\text{Recall}@k$ and gene Top- k accuracy. **P4: Family-aware differential diagnosis** adds pedigree or trio information and predicts the disease and mode of inheritance; it is deferred in v1. **P5: Clinical-communication faithfulness** evaluates whether the final answer is factual, relevant, supported by the available evidence, and consistent with the system trace. Automated P5 scores are calibrated against physician ratings on a 200-case subset, with inter-rater agreement reported using Cohen’s κ .

P1 and P5 are evaluated separately because neither is reducible to final diagnostic accuracy. Extraction errors alter the evidence available to downstream reasoning, while a correct diagnosis can still be supported by fabricated citations or an inconsistent rationale. Bias is therefore treated as a cross-cutting analysis rather than a separate pillar: every applicable metric is stratified by ancestry, disease prevalence, sex, age, language, and HPO density, following the multi-metric evaluation approach of HELM and MedHELM (Bedi et al. 2026).

Four-Layer Data Stack

The benchmark contains 84,450 instances from five sources, organized into four layers that vary evidence structure, clinical realism, scale, and contamination risk. Figure 2 summarizes the resulting capability–dataset evaluation surface; detailed disease counts, identifiers, modalities, and availability masks appear in Appendix A.

- **L1: Phenotype backbone (11,173 instances).** Phenopacket Store contributes 10,051 standardized cases with curated HPO terms and, when available, structured variants; RareBench-HF contributes 1,122 cases for comparison with established rare-disease LLM evaluation (Danis et al. 2025; Chen et al. 2024).
- **L2: Real-EHR noise (416 instances).** We construct a rare-disease probe from MIMIC-IV v3.1 discharge sum-

maries (?). Candidate admissions are identified through ICD-10–ORPHA mappings, with Orphadata categories marked as non-rare in Europe excluded (?), and are then passed through a de-leaking funnel that keeps only exact ICD–ORPHA maps, genuinely rare principal diagnoses, and cases whose narrative does not disclose the answer (“De-Leaking the EHR Layer” below). The agent sees the *presentation* segment only, truncated before the first diagnosis-revealing section with the gold disease name masked verbatim. This layer tests phenotype extraction and diagnosis under abbreviations, missingness, and longitudinal clinical documentation.

- **L3: Free-text scale (72,661 task instances).** RareArena contributes 49,760 screening instances and 22,901 confirmation instances derived from PubMed Central case reports (Chen et al. 2026a). We treat them as task instances rather than unique patients because the two tasks expose different evidence views.
- **L4: Temporal holdout.** From 2,401 post-cutoff PubMed Central Open Access candidates, we extract diagnoses and phenotypes, map them to Orphanet, and manually retain 200 cases satisfying four criteria: a definitive diagnosis, faithful phenotype extraction, verified publication after the prespecified cutoff, and confirmation that the target condition is rare. The extraction and adjudication protocol, including reviewer agreement, is provided in Section and Appendix D.

All layers are stratified where metadata permit by prevalence, clinical specialty, language, age, sex, ancestry, and HPO density. Dataset-specific licenses and access restrictions are documented in the released data cards; credentialed MIMIC-IV data are not redistributed.

Evaluation sample sizes. We distinguish the released benchmark pool from the cases executed in each experiment. RareBench-HF is evaluated in full for the primary LLM backbones and all applicable classical or offline baselines; the de-leaked MIMIC-IV note probe is evaluated on its full frozen 416-case set in all 24 agent–backbone cells. For Phenopacket Store and RareArena, each primary agent–backbone cell is evaluated on a prevalence-stratified random sample of 500 instances (seed 42), with proportional allocation across prevalence bands. DeepSeek V4-Pro and GPT-5 cells use 100–500 instances according to the preregistered compute budget; every result reports its denominator, and cells with fewer than 100 observations are explicitly marked. We do not extrapolate sampled results to the full pool. Instead, we report bootstrap confidence intervals per cell and verify that each 500-case sample reproduces the pool’s prevalence and HPO-organ-system distributions within two percentage points.

Canonical Case Representation

Every source is normalized into a single Pydantic-v2 record, `CanonicalCase`, from which each system adapter constructs the agent’s native input (Figure 3). The schema contains source identity and language; demographics; raw or synthetic free text; observed and excluded HPO terms; structured variants and local VCF pointers; family information;

parallel gold identifiers; and provenance metadata. Explicit availability masks prevent adapters from accessing evidence not assigned to the current capability or evaluation pass.

Three choices support auditable cross-dataset evaluation. First, gold diagnoses retain parallel OMIM, ORPHA, and source-specific identifiers, allowing ontology-aware matching without discarding the source label. Second, `free_text_vignette` and `synthetic_vignette` are distinct fields, so results based on generated prose remain identifiable. Third, `vcf_path` is a local-only pointer. Raw credentialed EHR text and local genomic files are never transmitted to external APIs; cloud adapters receive only the approved, de-identified abstraction required by the evaluated pillar.

Dual-Pass Evaluation

Where both curated HPO terms and source text are available, we evaluate two input conditions. **Pass A (gold HPO)** supplies the same curated phenotype set directly to each downstream diagnostic system, isolating reasoning and enabling controlled cross-system comparison. **Pass B (end to end)** supplies the raw clinical narrative and requires the system to extract phenotypes before diagnosis, approximating deployment. The Pass-A–Pass-B difference quantifies the downstream cost of phenotype extraction and turns input heterogeneity into a measured factor rather than an uncontrolled confound. This design extends RareBench’s comparison of structured phenotype and narrative inputs (Chen et al. 2024).

Metrics and Diagnostic Matching

Primary metrics follow the capability definitions above: precision, recall, and F1 for P1; Recall@1/3/5/10, MRR, and median rank for P2 and the disease component of P3; gene Top-*k* for P3; and the physician-calibrated four-axis score for P5. Across applicable tasks, we additionally report task success, pass^k reliability, calibration, latency, total cost, and cost-normalized accuracy (Yao et al. 2025). Results are accompanied by bootstrap confidence intervals and the multiplicity correction specified in the preregistration.

Disease predictions are matched hierarchically. We first test exact equality after identifier normalization, then accepted OMIM–ORPHA mappings from Orphadata, and finally canonical Orphanet names or synonyms. Fuzzy name matching is used only when no valid identifier is returned and requires a RapidFuzz score of at least 90. This threshold was selected from 217 manually audited borderline predictions; more than 85% of candidates scoring 70–89 were false positives. We release the mapping version, normalization rules, and complete audit trail.

Experimental Setup

The experimental setup is designed to separate backbone, scaffold, and implementation effects while preserving each system’s native workflow. We standardize model access and logging, isolate system dependencies through adapters, and freeze all confirmatory analyses before evaluating the temporal holdout.

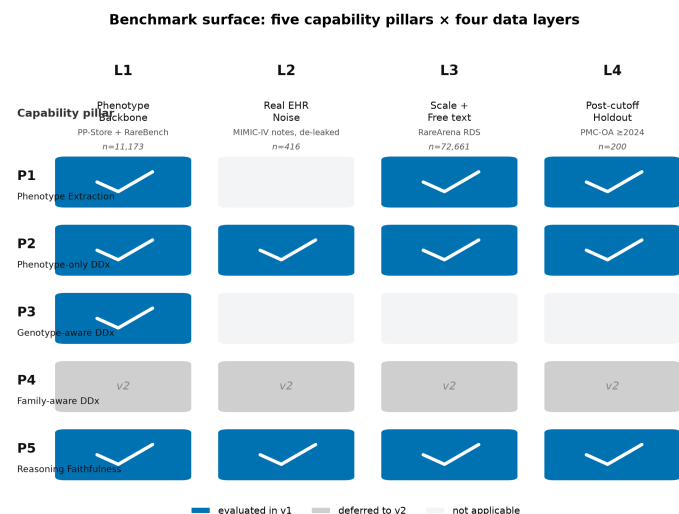


Figure 2: RareAgentBench evaluation surface. Rows denote five conceptual capability pillars and columns denote four data layers. Filled cells are evaluated in v1; grey cells, including family-aware diagnosis (P4), are reserved for future releases; light cells are not applicable.

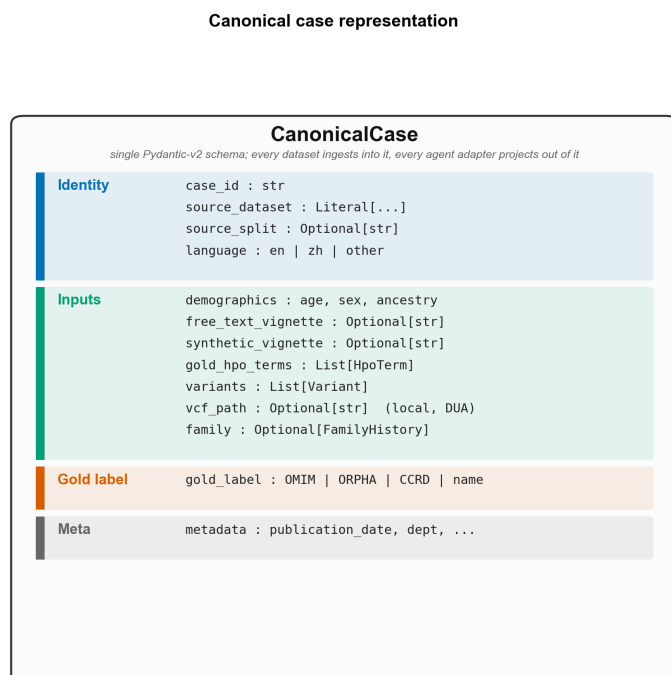


Figure 3: The CanonicalCase representation. Each dataset is normalized into a shared record, and isolated adapters project only the evidence permitted by the current capability and evaluation pass into each system’s native input.

Backbones

We evaluate each LLM-based system, where technically applicable, with four backbones spanning the observed cost-capability range: DeepSeek V4-Flash, DeepSeek V4-Pro, Gemini 3 Flash, and GPT-5. Gemini 3 Flash is the primary backbone; DeepSeek V4-Flash represents the low-cost setting, while V4-Pro and GPT-5 probe higher-capability regimes. Requests are routed through OpenRouter to provide a common API, billing surface, and generation record (?). The supplementary material (Appendix B) reports the immutable model identifier, provider route, access date, context window, token prices, and complete generation configuration for every evaluated cell. We do not use moving “latest” aliases.

Primary comparisons use the lowest reasoning setting reliably supported by each endpoint. This constrained regime serves three purposes: it reduces backbone-side test-time compute as a confound when measuring scaffold effects; approximates the operating conditions of systems originally developed before dedicated reasoning models (Tang et al. 2024; Kim et al. 2024; Schmidgall et al. 2026); and makes multi-call evaluation computationally tractable. OpenRouter normalizes reasoning controls across providers but does not guarantee equivalent latent computation across models (?). We therefore report the exact request parameters and observed reasoning-token usage rather than describing the configurations as computationally identical. Conclusions concerning GPT-5 and V4-Pro apply specifically to this constrained-reasoning regime. A reasoning-enabled single-LLM control is evaluated separately as preregistered ablation A8.

Pilot compatibility tests showed that hidden reasoning could exhaust the output budget of short-turn adapters, producing empty content, parser errors, or timeouts. For the frozen primary runs, GPT-5 uses `reasoning_effort=minimal`, and V4-Pro uses `reasoning.enabled=false`, the only tested configuration that consistently returned final-answer content within the adapter limits. These settings are passed unchanged into isolated subprocesses and verified from request and token-usage receipts. Root-cause experiments, including attempted token caps and latency measurements, are reported in Appendix B rather than treated as benchmark results.

Failure handling is fixed before aggregation. Empty outputs, parser errors, and timeouts count as unsuccessful attempts and remain in the denominator. A cell is marked unavailable only when the adapter-backbone combination cannot execute the system’s required interaction protocol after the documented compatibility procedure. MAI-DxO with GPT-5 is the sole such cell in the current matrix and is reported as missing, not silently removed. Every table reports the attempted and successful sample counts.

System Adapters

The eight diagnostic systems have mutually incompatible dependencies and, in several cases, different language-model SDK generations. We therefore execute each system in an isolated runtime behind a common AgentAdapter interface. An adapter receives a CanonicalCase (Section),

exposes only the evidence permitted by the current capability and evaluation pass, converts it to the system’s native input, invokes the isolated process, and returns a `PredictionLog`. The log records ranked diagnoses, extracted HPO terms, reasoning or communication traces when available, token usage, cost, latency, and a normalized status code.

Adapter modifications are restricted to transport, input projection, output parsing, and instrumentation. We do not alter prompts, tool-selection policies, stopping rules, retrieval corpora, or orchestration logic unless a change is required to replace an unavailable endpoint; any such deviation is declared in the agent fairness matrix (supplementary material, Appendix C). For OpenRouter-compatible systems, the patch changes only client construction and forwards the model and reasoning configuration through the subprocess environment. Per-system patches and immutable environment specifications are released with the benchmark.

This design provides three audit properties. First, every run receipt records the canonical case identifier, adapter version, model identifier, configuration hash, subprocess command, random seed, and termination status. Second, a common logging layer aggregates per-call token usage, latency, and billed cost into a case-level receipt. API costs are exact when returned by the gateway; calls that bypass the wrapper are estimated from recorded token counts and versioned prices and are labeled accordingly. Offline methods have zero API cost, with compute time reported separately. Third, process isolation contains dependency, parser, retrieval, and state failures within the affected system.

Adapter overhead is measured independently and reported alongside end-to-end latency; interpreter startup adds approximately 0.5–2 seconds per case. We also clear system-specific output directories between cases to prevent cross-case state leakage. These safeguards, including the three upstream systems that required explicit state cleanup, are documented in Appendix B.

Preregistration and Holdout Governance

Before temporal-holdout unblinding, we freeze H1–H11, ablations A1–A12, the staged sampling budget, cell-level stopping rules, primary metrics, the Holm–Bonferroni correction, and the LLM-judge agreement protocol. The frozen protocol, timestamped repository commit, and anonymized read-only OSF registration are supplied with the supplementary material. This design responds to documented contamination and reproducibility risks in rare-disease LLM evaluation (Nguyen et al. 2026).

The 200-case PMC Open Access temporal holdout is evaluated once under the frozen agent set, matching rules, and analysis plan. Phenopacket Store, RareBench, RareArena, and the de-leaked MIMIC-IV note probe remain development layers, but no holdout result is used to revise prompts, adapters, thresholds, or sampling decisions. Confirmatory cells ship with a reproducibility receipt containing the run identifier, request identifiers when available, configuration hash, denominator, failures, and cost. Exploratory analyses are labeled explicitly and are not included in confirmatory hypothesis testing.

Main Results

We report variant-aware Recall@1 (R@1) across agents, backbones, and datasets, followed by backbone sensitivity and cost–accuracy analysis. Complete hypothesis tests, confidence intervals, and execution receipts appear in the supplement.

Headline R@1 Matrix

The average is reported only for LLM configurations covering all four development datasets and is not comparable to the two-dataset coverage of LIRICAL and VC-RDAGENT. DeepRare with V4-Pro reaches .44 on 36 RareBench P3 cases through the HPO-plus-variant channel, but the sample is too small for the primary ranking. All V4-Pro cells use the reasoning-disabled setting in Section . MAI-DxO with GPT-5 is unavailable on the HPO-input layers under the frozen adapter protocol, but all four MAI-DxO backbones do complete the note probe; the failure analysis is in the supplement. The MIMIC-N column supersedes the leaked MIMIC-IV numbers of the earlier draft, which were computed on notes whose concatenated ICD-10 titles contained the gold answer verbatim; the two are not cell-by-cell comparable because the case set changed as well (“De-Leaking the EHR Layer” below). The PMC-OA column is reported for completeness at 8 of 24 cells because the holdout is unblinded once and cannot be extended without spending it; it is read as a within-layer pre- versus post-cutoff comparison (supplement, “Temporal-Holdout Protocol”) rather than as a cross-agent ranking, and its higher absolute level reflects the diagnostic legibility of published case reports rather than easier diseases.

De-Leaking the EHR Layer

The MIMIC-N column required a dedicated construct-validity audit, because a rare-disease cohort assembled from billing codes leaks its own answer. Each filter in the funnel removes a distinct threat: 150,033 admissions with any ICD–ORPHA mapping → 107,144 under *exact* mappings only (dropping narrower-to-broader and broader-to-narrower approximate maps) → 18,392 whose gold code is a genuinely rare disease → 2,325 where that disease is the admission’s *principal* diagnosis rather than comorbidity or past history → 1,255 carrying a usable discharge summary (117 diseases) → 1,083 after dropping cases whose narrative states the diagnosis as already known → 855 in which the gold disease name never appears verbatim → 687 after a strict synonym screen (105 diseases) → 416 on the HPO-annotated subset (68 diseases), which is the frozen scored set.

Two of these steps are anti-leakage rather than cohort definition. First, a ± 45 -character window around each masked diagnosis detects history cues (“h/o”, “s/p”, “known”), removing 172 cases in which the note frames the target as an established diagnosis. Second, masking the gold disease *name* is insufficient: screening against the full Orphanet synonym and abbreviation list shows that 19.6% of the nominally clean cases still spelled the answer out under an alternative label (eponyms and all-capitals abbreviations such as ADPKD or GBS, which the crossmap-aware scorer credits as hits), and removing them costs a further 168 cases.

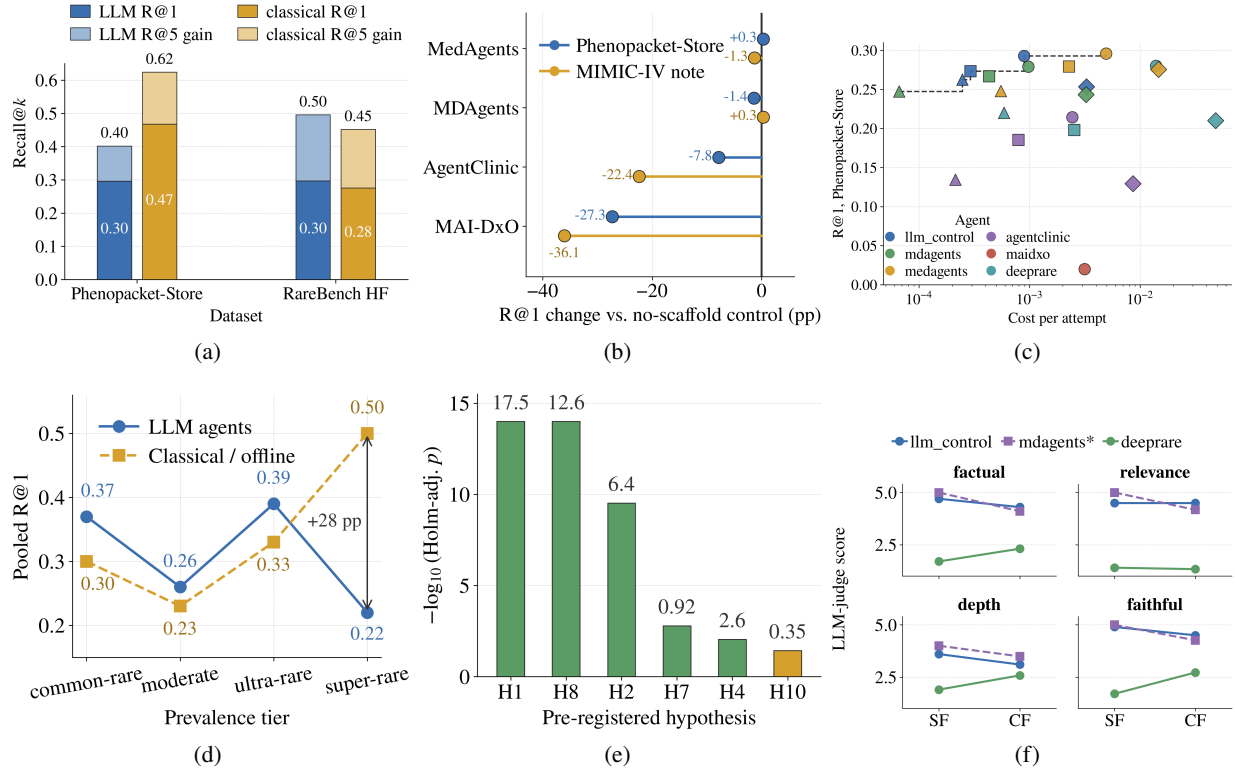


Figure 4: Main results and analysis. Recall is variant-aware, over the attempted denominator. **(a)** Best LLM agent vs. best classical/offline baseline on the two HPO-input datasets where a classical run is possible. **(b)** Each scaffold’s R@1 change (pp) vs. the no-scaffold single-LLM control; the MIMIC series is the de-leaked note probe (mean over four backbones, 416 cases). **(c)** Cost per attempt (USD, log) vs. R@1 for every executable agent–backbone cell on Phenopacket-Store; shape = backbone (circle Gemini 3 Flash, square V4-Pro reasoning-off, triangle V4-Flash, diamond GPT-5 minimal), dashed step = Pareto frontier. **(d)** Pooled R@1 by Orphanet prevalence tier; the families cross on the rarest tier, where training-frequency priors are weakest. **(e)** Pre-registered tests under Holm–Bonferroni, as $-\log_{10}$ of the Holm-adjusted p (bars truncated at the axis top; printed number = test statistic, z for H1/H8/H2/H4, Spearman ρ for H7/H10). Green survives at $\alpha=.05$, gold does not. H1: classical > LLM, super-rare; H8: R@1 peaks at 16–30 HPO terms; H2: genotype-channel lift; H7: specialty-difficulty correlation; H4: scaffolding helps complex cases; H10: faithfulness–accuracy decoupling. **(f)** Pillar-5 trace scores under a judge swap on identical traces: same-family (SF: Gemini 3 Flash) vs. cross-family (CF: Claude Sonnet 4.5). *mdagents** also had its v2 trace repaired, so its SF→CF change mixes judge swap with better coverage; *llm_control*/*deeprare* traces were complete and isolate the judge effect.

Table 2 quantifies what this buys, holding the case set fixed: the leaked variant inflates R@1 by 6–20 points and, more damagingly, changes the ranking. Under leakage V4-Flash appears to be the strongest backbone by 5 points; on the same cases with the answer removed, all four backbones collapse into a .353–.373 band that is within bootstrap noise. A backbone recommendation read off the leaked slice would therefore have been an artifact of which model is best at copying a diagnosis string out of its own input. We report the de-leaked column only, and treat the leaked one as a measurement of the leak.

Two caveats remain. The 416-case set is restricted to the 68 diseases with HPO annotations available in our mapping, dropping 271 otherwise-valid cases across 37 diseases; the loss is an annotation-coverage artifact and not related to difficulty, but it does narrow disease diversity. And the screen

operates at the case level: a disease may still be inferable from a characteristic treatment or laboratory pattern that no name-based filter can catch.

Backbone–Scaffold Interaction

Table 3 summarizes PP-Store sensitivity to backbone choice using the cells in Table 1.

Gemini is strongest on PP-Store for every system family, with narrow spreads for the LLM control, MDAgents, and MedAgents (.03–.04) but larger effects for AgentClinic (.08) and DeepRare (.06). The ordering changes elsewhere: on the de-leaked MIMIC-IV note probe the four backbones are statistically indistinguishable for the unscaffolded control (.353–.373, a .02 spread), GPT-5 minimal is by far the strongest for MDAgents (.435 vs. .317–.368), and V4-Pro is strongest for AgentClinic (.216, against .048 for GPT-5 min-

imal). Backbone effects are therefore scaffold- and dataset-dependent. Because several bootstrap intervals overlap, we do not treat small gaps as definitive rankings.

Cost–Accuracy Trade-offs

The evaluation comprises 68,668 predictions at a total API cost of \$191.76, or \$0.0028 per prediction including offline runs. Per-prediction cost spans more than an order of magnitude: DeepSeek V4-Flash is cheapest (\$0.00040), then V4-Pro reasoning-off (\$0.00088), Gemini Flash (\$0.00321), and GPT-5 minimal the most expensive (\$0.00793); the classical/offline baselines incur no API cost. Relative to V4-Flash, V4-Pro costs 2.2× more, Gemini 8× more, and GPT-5 minimal 19.8× more; GPT-5 is also 9× more expensive than V4-Pro. As Figure 4?? shows, this cost ordering does not induce a matching accuracy ordering. The evaluation used 53% of the preregistered \$360 budget; per-backbone spend and per-cell accounting are in Appendix J.

Headline Findings

The final matrix incorporates two corrections: V4-Pro was rerun with reasoning disabled after hidden-reasoning output starvation, and PP-Store/RareArena runs were harmonized to a shared 2,000-case sampling frame where technically feasible. R@1 decreased by approximately 2–4 points relative to earlier 500-case estimates, strengthening the structured-baseline result and removing the earlier claim that GPT-5 is the best MedAgents backbone.

F1: Classical and offline methods outperform scaffolded LLMs on HPO input. On PP-Store, LIRICAL reaches .47 R@1 and VC-RDAGENT .44, compared with .30 for the best scaffolded LLM cell (Figure 4??). The resulting 17-point gap between LIRICAL and MedAgents persists on the larger harmonized sample. RareBench shows a related pattern: DeepRare (.29–.30), VC-RDAGENT (.28), and LIRICAL (.23) lead, while all other LLM scaffolds remain at or below .10. The magnitude of the RareBench gap is partly attributable to ontology alignment, as examined under F5.

F2: Multi-agent scaffolding does not reliably beat a single-LLM control. The same pattern holds on two independent datasets (Figure 4??): Phenopacket-Store (Gemini Flash) and the de-leaked MIMIC-IV note probe (mean over four backbones). No scaffold clears the no-scaffold control by a meaningful margin—MedAgents and MDAGENTS stay within 1.4 points of the control on both datasets—while the heavier-orchestration scaffolds fall well below it: AgentClinic –7.8 (PP-Store) and –22.4 (MIMIC), MAI-DxO –27.3 and –36.1. On the de-leaked MIMIC matrix the unscaffolded control (mean R@1 .364) is statistically tied with the best light scaffolds (MedAgents .350, MDAGENTS .367, overlapping CIs) and far above the heavy ones (AgentClinic .140, DeepRare .011, MAI-DxO). The evidence thus supports a conditional, often negative scaffold effect—heavy orchestration consistently hurts—rather than a general multi-agent advantage.

F3: V4-Flash is cost efficient but not quality equivalent to Gemini. V4-Flash costs \$0.00040 per prediction,

compared with \$0.00321 for Gemini. On PP-Store, it trails Gemini by 3 points for the LLM control and MDAGENTS, 4 for MedAgents, and 7 for AgentClinic. On the de-leaked MIMIC-IV note probe the sign flips for the light-coordination scaffolds—V4-Flash is 2 points *below* Gemini for the LLM control and 6 below for MedAgents, but 5 points *above* it for MDAGENTS and 4 above for AgentClinic. V4-Flash also exhibits more transient empty-content responses on free-text and HPO-list inputs, mitigated by the frozen wrapper-level retry policy. The appropriate conclusion is therefore lower cost with an agent- and modality-dependent accuracy trade-off, not quality parity.

F4: GPT-5 minimal reasoning carries the highest cost without a consistent accuracy advantage. GPT-5 minimal is competitive in selected cells—it holds the single best de-leaked MIMIC-IV cell in the whole matrix (MDAGENTS, .435)—but it is not the sole best backbone for any scaffold on PP-Store, and it is the *worst* of the four backbones for AgentClinic on both PP-Store (.13) and the note probe (.048). At 19.8× the per-prediction cost of V4-Flash, these isolated gains do not establish a favorable overall trade-off. The preregistered H6 ablation isolates the reasoning channel: enabling reasoning for the V4-Pro single-LLM control changes R@1 by only +.008, while yielding no parseable answer in 40% of cases and increasing latency by 10–40×. These findings apply to the evaluated adapters and token limits rather than to unconstrained GPT-5 quality in general.

F5: RareBench exposes an evaluator–ontology interaction. RareBench is uniquely difficult for general LLM scaffolds but remains tractable for systems operating directly over HPO and disease ontologies. Error analysis identifies sibling-level ORPHA distinctions and incomplete OMIM–Orphanet mappings as recurrent failure modes. For example, Methylmalonic acidemia with homocystinuria (ORPHA:26) and Vitamin B12-unresponsive methylmalonic acidemia (ORPHA:27) are clinically related but receive different exact identifiers. Classical and HPO-centered pipelines can bypass part of this mismatch through OMIM identifiers and Orphanet name matching. Variant-aware fuzzy matching recovers 1–8 R@1 points across affected cells but does not close the performance gap. We therefore report both exact and variant-aware scores and treat the result as an interaction among model behavior, cross-ontology mapping, and evaluator policy rather than as a pure model failure.

Prevalence and pre-registered tests. Stratifying by Orphanet prevalence tier sharpens F1 (Figure 4??). LLM agents peak on common and ultra-rare tiers but fall to R@1 .22 on super-rare disease, whereas the classical/offline baselines *rise* to .50 there—a 28-point reversal on exactly the rarest cases. Across the pre-registered family (Figure 4??), five of the six testable hypotheses survive Holm–Bonferroni correction, including the classical-over-LLM super-rare gap (H1) and the genotype-channel lift (H2); only the faithfulness–accuracy decoupling (H10) fails and is reported as exploratory (Section).

Self-Preference Bias in LLM-as-Judge

Protocol. We evaluate Pillar 5 reasoning traces on four five-point axes: factuality, relevance, depth, and faithfulness. The initial protocol used Gemini 3 Flash Preview as the judge while evaluating agents instantiated with the same backbone family. We then rescored the traces with Claude Sonnet 4.5 using the same rubric and judge prompt (Figure 4??). Prior work shows that LLM evaluators can favor generations from their own model family, motivating an explicit judge-family sensitivity analysis (?). Because several v2 runs also repair trace-capture failures, however, the complete v1–v2 difference is not a clean causal estimate of self-preference. We therefore distinguish fixed-trace judge sensitivity from changes induced by improved trace coverage.

Judge-family sensitivity. For the fixed `llm_control` traces, replacing Gemini with Claude reduces factuality, depth, and faithfulness by .40–.50 points while leaving relevance unchanged. This directional shift is consistent with same-family preference, but a two-judge comparison cannot separate self-preference from judge-specific calibration without human labels or a reciprocal design. The cross-agent comparisons are additionally affected by trace coverage. Under Claude, the `llm_control` margins over MDAgents are $\{+.20, +.33, -.39, +.24\}$ across factuality, relevance, depth, and faithfulness: MDAgents leads on depth, whereas the control remains slightly ahead on the other axes. The remaining margins lie within the bootstrap intervals reported in Appendix E.

After collapsing scores into $\{1-2, 3, 4-5\}$, inter-judge agreement is $\kappa = .62$. Given the 10-case-per-agent pilot sample, we treat the per-axis deltas and ranking changes as exploratory. The principal result is that judge choice materially affects the interpretation of trace quality, not a precise estimate of a universal self-preference effect.

Trace-capture failures. The v2 pipeline also fixes two independent instrumentation errors. MAI-DxO’s panel `conversation_history` was not passed to the judge in v1, producing 10/10 JSON-parse failures; the corrected pipeline captures 26,972 characters and eliminates these failures. MDAgents previously exposed only the 337-character moderator verdict rather than the full deliberation. Capturing the complete 20,034-character multi-expert trace reduces judge errors from 8/10 to 0. These fixes are necessary for valid process evaluation, but they also mean that the affected v1–v2 score differences cannot be attributed solely to judge family. The patches and replay receipts are documented in Appendix B.

Effect on H10. Judge choice also bears on the preregistered H10 verdict, but only once trace capture is controlled. An earlier analysis compared $\rho = .098$ (Gemini) against $\rho = .616$ (Claude) between judged faithfulness and top-1 accuracy and read this as a large judge-family gap; that split was a trace-capture artifact, because the Gemini scores had been computed on truncated or empty traces while the Claude scores used the repaired traces. Re-running the Gemini judge on the *same* repaired traces ($n=40$) raises its correlation from .098 to $\rho = .457$, against Claude’s $\rho = .640$,

and the two judges’ faithfulness scores themselves agree at $\rho = .741$. The residual judge-family difference is therefore modest, and both estimates sit near the preregistered $\rho < .5$ decision boundary—one just below, one just above—so we report H10 as judge-dependent and exploratory rather than as a headline confirmation and withdraw the earlier .098-vs-.616 contrast. The durable point is that judge family and trace-capture completeness are both first-order evaluation factors that must be controlled together.

Implications for benchmark design. LLM-based evaluation is increasingly used for open-ended medical and agent outputs, including MedHELM’s multi-model jury and MedR-Bench’s automated reasoning evaluator (Bedi et al. 2026; ?). Our results reinforce prior evidence that a heterogeneous judge panel can reduce intra-model bias relative to a single judge (?). Ablation A12 therefore evaluates the same predictions using exact OMIM/ORPHA matching, BioLORD-assisted synonym matching, a GPT-5 judge, and physician adjudication on 200 cases.

For benchmark builders, deterministic scoring should remain primary whenever a verifiable target exists. Subjective trace evaluation should replay identical frozen traces across judges from different model families, report judge-conditioned results or panel consensus, and include a human-audited subset for consequential claims. We also record the effect of every judge substitution rather than merging scores across judge versions. For traces exceeding 5,000 characters, our implementation uses overlapping 3,000-character windows with a 500-character overlap and averages per-axis scores, preventing silent truncation of traces such as DeepRare’s 21,401-character output. The complete unchunked trace remains available for audit.

Table 1: Variant-aware R@1 for each reported agent–backbone configuration across all five data sources. Rows are grouped by backbone, with the same six agents in the same order inside every block, so a column read down one block is a scaffold comparison at fixed backbone and the same row position across blocks is a backbone comparison at fixed scaffold. Classical and offline systems are listed first; their free-text cells are not applicable because they require structured HPO input. **MIMIC-N** is the de-leaked discharge-note probe (“De-Leaking the EHR Layer” below): all 24 agent–backbone cells are complete on the same frozen 416-case set, with the denominator fixed at 416 so that every failure, timeout, parser error, and empty-but-successful response counts as a miss. **PMC-OA** is the L4 temporal holdout, which is unblinded exactly once and was run on two backbones for the four LLM-scaffold systems only; *n/r* marks the 16 cells that were never run, and nothing is imputed for them. This is a deviation from the pre-registered H3 design, which named the retrieval-capable agents: DeepRare and MAI-DxO were dropped because both score at or near zero on the comparable free-text MIMIC-N probe, and V4-Pro and GPT-5 entered the lineup after the unblinding (supplement, “Temporal-Holdout Protocol”). Per-cell attempted counts for every cell in this table are listed in the supplement (“Per-Cell Sample Counts”); they vary widely, and the DeepRare and MAI-DxO rows in particular rest on far smaller samples than the scaffold rows, so those two rows should be read against that table. Published case reports are written to be diagnostically legible, so the absolute PMC-OA level is not comparable to the other layers and is excluded from Avg.; the load-bearing holdout analysis is the difficulty-matched pre- versus post-cutoff contrast in the supplement, not this column. **Avg.** is the mean over the four development layers. Cell shading encodes R@1: <.10 .10–.20 .20–.30 .30–.40 ≥.40.

Backbone	Agent	PP-Store	RareAren	Bench	MIMIC-N	PMC-OA	Avg.
<i>Classical / offline (structured HPO input only)</i>							
—	LIRICAL	0.47	n/a	0.23	n/a	n/a	n/a
—	VC-RDAgent	0.44	n/a	0.28	n/a	n/a	n/a
Gemini 3 Flash	LLM control	0.29	0.28	0.02	0.37	0.62	0.24
	MDAgents	0.28	0.28	0.10	0.32	0.62	0.24
	MedAgents	0.30	0.30	0.05	0.40	0.63	0.26
	AgentClinic	0.21	0.14	0.01	0.13	0.28	0.12
	DeepRare	0.28	0.00	0.30	0.01	n/r	0.15
	MAI-DxO	0.03	0.07	0.01	0.00	n/r	0.03
DeepSeek V4-Pro	LLM control	0.27	0.19	0.02	0.37	n/r	0.21
	MDAgents	0.27	0.22	0.04	0.35	n/r	0.22
	MedAgents	0.28	0.23	0.01	0.35	n/r	0.22
	AgentClinic	0.18	0.12	0.01	0.22	n/r	0.13
	DeepRare	0.25	0.00	0.44	0.01	n/r	0.18
	MAI-DxO	0.00	0.00	0.00	0.00	n/r	0.00
DeepSeek V4-Flash	LLM control	0.26	0.21	0.05	0.35	0.48	0.22
	MDAgents	0.25	0.23	0.05	0.37	0.53	0.22
	MedAgents	0.26	0.24	0.05	0.34	0.36	0.22
	AgentClinic	0.14	0.11	0.02	0.17	0.25	0.11
	DeepRare	0.22	0.00	0.29	0.02	n/r	0.13
	MAI-DxO	0.00	—	—	0.00	n/r	n/a (2-ds)
GPT-5 minimal	LLM control	0.26	0.22	0.01	0.36	n/r	0.21
	MDAgents	0.24	0.23	0.01	0.44	n/r	0.23
	MedAgents	0.28	0.26	0.01	0.31	n/r	0.22
	AgentClinic	0.13	0.10	0.00	0.05	n/r	0.07
	DeepRare	0.22	0.00	0.21	0.00	n/r	0.11
	MAI-DxO	—	—	—	0.00	n/r	n/a (1-ds)

Table 2: Leakage magnitude on *identical* cases: unscaffolded LLM-control R@1 before and after de-leaking, same 416 admissions. Leakage both inflates accuracy by 9–20 points and reorders the backbones—V4-Flash looks best when leaked but is mid-field once the answer is removed.

Backbone	Leaked	De-leaked	Δ
DS V4-Flash	.553	.353	−.200
DS V4-Pro	.500	.365	−.135
Gemini Flash	.452	.373	−.079
GPT-5 minimal	.425	.363	−.062

Table 3: Per-agent backbone sensitivity on PP-Store. The range is the difference between the best and worst reported backbone for that system.

Agent	Best backbone	Worst backbone	R@1 range
LLM control	Gemini Flash (.29)	DS V4-Flash / GPT-5 minimal (.26)	.03
MDAgents	Gemini Flash (.28)	GPT-5 minimal (.24)	.04
MedAgents	Gemini Flash (.30)	DS V4-Flash (.26)	.04
AgentClinic	Gemini Flash (.21)	GPT-5 minimal (.13)	.08
DeepRare	Gemini Flash (.28)	DS V4-Flash (.22)	.06

Table 4: Reasoning-trace scores under the Gemini-family judge (v1) and Claude judge (v2), reported as $v1 \rightarrow v2$. Each agent uses 10 stratified Phenopacket-Store cases (seed 42). The rubric and judge prompt are unchanged, but trace-capture fixes affect MDAgents and MAI-DxO and slightly change the captured DeepRare trace length; their score differences therefore combine judge and instrumentation effects.

Agent	Factuality	Relevance	Depth	Faithfulness	Trace length	Main change
llm_control	4.70 $\rightarrow$ 4.30 ($-.40$)	4.50 $\rightarrow$ 4.50	3.60 $\rightarrow$ 3.10 ($-.50$)	4.90 $\rightarrow$ 4.50 ($-.40$)	986 characters	Three axes decrease; relevance is unchanged
mdagents	5.00 $\rightarrow$ 4.10	5.00 $\rightarrow$ 4.17	4.00 $\rightarrow$ 3.49	5.00 $\rightarrow$ 4.26	337 $\rightarrow$ 20,034	Under Claude, depth exceeds llm_control (3.49 > 3.10)
deeprare	1.70 $\rightarrow$ 2.31 ($+.61$)	1.40 $\rightarrow$ 1.33 ($-.07$)	1.90 $\rightarrow$ 2.58 ($+.68$)	1.70 $\rightarrow$ 2.72 ($+1.02$)	18,429 $\rightarrow$ 21,401	Three axes improve; relevance is nearly unchanged
maidxo	n/a $\rightarrow$ 2.11	n/a $\rightarrow$ 1.85	n/a $\rightarrow$ 1.64	n/a $\rightarrow$ 1.88	0 $\rightarrow$ 26,972	v1: 10/10 parse failures; v2: 0/10

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